

ZHIYU WANG

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Agents LLMs AI4Bio/Health Geometric DL



🏛️ EDUCATION

University of Cambridge

Advanced Computer Science, Master of Philosophy (Distinction, Rank: 8/60)

Cambridge, UK

Oct 2024 — Jul 2025

– Thesis awarded the Highly Commended MPhil Project Prize 2024–2025 by the Department of Computer Science and Technology

University College London

Computer Science, Bachelor of Science (First Class Honors)

London, UK

Sep 2021 — Jun 2024

🔬 RESEARCH EXPERIENCE

Auto-Research for the Virtual Cell Challenge | SAPIENT INTELLIGENCE & SJTU

Remote

Project lead | *Collaborator: Prof. Liang Hong*

Aug 2026 — Present

– Leading a joint project with Shanghai Jiao Tong University (SJTU) that applies Praxist, our AI-scientist framework, to zero-shot prediction of CRISPRi knockdown responses in the [Virtual Cell Challenge 2026](#); ranked 167/1,169 (top 15%) on the preliminary leaderboard.

Bayesian Optimization for Auto-Research Systems | MILA – QUEBEC AI INSTITUTE

Remote

Collaborator | *Supervisor: Prof. Jian Tang*

Apr 2026 — Present

– Integrated Bayesian optimization into agentic evolution: a Gaussian process predicts gains; Thompson sampling picks the next experiment.
– Achieved the best average rank on 15 AlphaEvolve tasks and the top gold-medal rate (45%) on MLE-bench Lite.

Protein Substructure Alignment via Optimal Transport | UNIVERSITY OF CAMBRIDGE & SJTU

Cambridge, UK

Project lead | *Supervisor: Prof. Pietro Liò & Prof. Liang Hong*

Apr 2025 — Sep 2025

– Developed PLASMA, a plug-and-play optimal-transport module for local substructure alignment on any protein language model. ([ICLR26](#))
– In detecting shared substructures, PLASMA reached ROC-AUC 0.96–0.99 vs. 0.81–0.93 for the widely used global aligner TM-align at $\sim 50\times$ its speed, and outperformed the state-of-the-art local method EBA at $\sim 3\times$ its speed, with much cleaner alignments.
– Showed with PyMOL case studies that the aligned substructures are structurally similar, not artifacts of sequence similarity.

Hierarchical Protein Representation Learning | UNIVERSITY OF CAMBRIDGE

Cambridge, UK

Master's thesis | *Supervisor: Prof. Pietro Liò*

Oct 2024 — Jul 2025

– Built a complete SE(3)-equivariant framework that adds a secondary-structure level to protein representations via a hierarchical hypergraph, allowing a coarser-grained representation that gives better global understanding. ([Invited Talk, Cambridge](#)) ([arXiv](#))
– Leveraged this new structural hierarchy to achieve the best SCOP fold classification accuracy (43.3% vs. 38.4% for the same network without it; 40.1% best baseline), confirming the improved global understanding of protein structure.

Multi-Omics Data Integration with GNNs | UNIVERSITY OF CAMBRIDGE

Cambridge, UK

Project lead | *Supervisor: Prof. Pietro Liò*

Feb 2025 — Sep 2025

– Built MoRE-GNN, a heterogeneous graph autoencoder learning relational graphs from single-cell RNA, protein and ATAC data; outperforms MOJITO on strongly correlated CITE-seq data (BM-CITE ARI 0.892 vs 0.868) and enables cross-modal prediction. ([arXiv](#))

Graph-based Pain Estimation from Facial Expressions | UNIVERSITY OF CAMBRIDGE

Cambridge, UK

Project lead | *Supervisor: Prof. Hatice Gunes*

Oct 2024 — Jan 2025

– Proposed GraphAU-Pain, combining a ResNet-50 backbone with a facial action unit (AU) graph for interpretable pain estimation; transfer learning for domain shift and missing labels raised average AU F1 on UNBC from 20% to 63%. ([IJCAI25 Workshop](#)) ([IEEE CEM](#))

Readmission Prediction with Remote Patient Monitoring Data | UNIVERSITY COLLEGE LONDON

London, UK

Undergraduate thesis | *Supervisor: Prof. Ivana Drobnjak*

May 2023 — Jun 2024

– Built readmission-prediction models (ROC-AUC 0.79 $\pm$ 0.03) from remote patient monitoring data with UCL Hospital clinicians, using SHAP to show which clinical features drive each prediction and patient-independent bootstrapping to cope with only 55 patients.

INDUSTRIAL EXPERIENCE

Autonomous Research Agents | SAPIENT INTELLIGENCE

ML Research Scientist

Beijing, China

Sep 2025 — Present

- Co-first author of **Praxist**, an autonomous AI-scientist framework with DAG-based execution lineage; on the 75-task MLE-bench, it won 60 medals (49 gold) vs. 55 (34 gold) for Claude Code with Opus, at about 1/12 of the cost.
- Showcased Praxist at **AI4 2026** through live booth demos and conversations with prospective customers and collaborators.

Healthcare LLM Knowledge Distillation | IFLYTEK HEALTHCARE

Intern ML Researcher

Hefei, China

Jul 2024 — Sep 2024

- Distilled a teacher LLM into synthetic medical instruction data to fine-tune the model behind iFLYTEK's **Xiaoyi** AI health-assistant app, raising instruction-following accuracy from 34% to 82%.

RAG-Based Financial Intelligence System | LUOJIN DATA INFORMATION

Intern ML Engineer

Shanghai, China

May 2023 — Aug 2023

- Built a RAG pipeline (MongoDB, LangChain) for financial report classification with 97% entity matching.

Homepage Navigation Assistant for Hospitals | UCL & NATIONAL HEALTH SERVICE

Industrial Collaboration Program

London, UK

Sep 2022 — May 2023

- Built a system that crawls a hospital's website and auto-generates an assistant that answers visitor questions and guides them to the right pages, cutting setup from weeks of manual development to 30 minutes; demoed at Great Ormond Street Hospital ([media coverage](#)).

PUBLICATIONS

Conference, Journal & Workshop Papers

* equal contribution

1. “Fast and Interpretable Protein Substructure Alignment via Optimal Transport.”

Zhiyu Wang, Bingxin Zhou, Jing Wang, Yang Tan, Weishu Zhao, Pietro Liò, Liang Hong. [ICLR 2026](#).

2. “Data-Efficient AU-Graph Transfer Learning for Consumer-Camera Pain-Aware Sensing.”

Zhiyu Wang, Han Wei Wang, Hangqian Li, Jun Jiang, Hatice Gunes, Yang Liu. [IEEE Consumer Electronics Magazine 2026](#).

3. “GraphAU-Pain: Graph-based Action Unit Representation for Pain Intensity Estimation.”

Zhiyu Wang, Yang Liu, Hatice Gunes. [IJCAI 2025 MiGA Workshop](#).

Preprints

4. “Praxist: From Experimental Artifacts to Solution Lineages.”

Jin Li*, Ahmed Murtadha*, **Zhiyu Wang***, Qiwen Chen*, William Chen*, Yifei Wu*, Guan Wang*, et al. [arXiv 2026](#).

5. “Topotein: Topological Deep Learning for Protein Representation Learning.”

Zhiyu Wang, Arian Jamasb, Mustafa Hajj, Alex Morehead, Luke Braithwaite, Pietro Liò. [arXiv 2025](#).

6. “MoRE-GNN: Multi-omics Data Integration with a Heterogeneous Graph Autoencoder.”

Zhiyu Wang, Sonia Koszut, Pietro Liò, Francesco Ceccarelli. [arXiv 2025](#).

ADDITIONAL EXPERIENCE

- **Reviewer Experience:** NeurIPS 2026; AI4Science Workshop at NeurIPS 2025 and ICML 2026
- **Teaching Experience:** Programming Tutor at UCL (2022–23)